Plagiarism Checker For Short Text

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PlagioScan

PLAGIARISM CHECKER FOR SHORT TEXT

TF-IDF VECTORIZATION + COSINE SIMILARITY

The Problem

! Existing tools miss short text
Turnitin, Grammarly built for long docs only

! Paraphrasing evades detection
Reworded content slips through basic checks

! No tool for quiz / tweet / blog text
Short-form content is a growing blind spot

Our Solution

PlagioScan

Built for 10-1000 word texts

+ Instant similarity percentage
+ RED / ORANGE / GREEN badge
+ Common keyword highlighting
+ No install - runs in browser

Algorithm and Approach

01 Input Text — Two short texts
→ **02** Preprocess — Clean and tokenize
→ **03** TF-IDF — Vectorize text
→ **04** Cosine Score — Measure angle
→ **05** Result — Score + Badge

```
cos(A,B) = (A dot B) / (||A|| x ||B||) | 75%+ HIGH . 45-74% MODERATE . below 45% LOW
```

Target Audience

- **Students and** Teachers
- **Academic** Institutions
- **Content** Writers
- **E-Learning** Platforms

Sales Pitch

+ No rival tool for short text plagiarism detection
+ Freemium: 5 checks/day free - upgrade from $9/month
+ GDPR-safe, API-ready, LMS integration in minutes

FREE
5 checks/day

$9/mo Basic $29/mo Pro
$49/mo Enterprise

PLAGIOSCAN - CATCH WHAT OTHERS MISS

1. Introduction

In today's digital world, copying content from the internet has become very common. Many students, writers, and professionals sometimes reuse text without proper citation, which leads to plagiarism. Plagiarism is a serious issue in academic and professional environments because it affects originality and credibility.

To address this problem, We developed a project named **PlagioScan**, which is a plagiarism detection system. This system compares two pieces of text and determines how similar they are. The main purpose of this project is to help users identify whether content is original or copied.

PlagioScan is designed to be simple, efficient, and user-friendly. It allows users to input two texts and instantly get a similarity score along with a detailed explanation. This makes it useful for students, teachers, and content creators.

2. Objectives of the Project

1. To detect similarity between two text inputs
2. To provide a percentage score of similarity
3. To classify the similarity level (Low, Moderate, High)
4. To highlight common keywords between the texts
5. To create a simple web-based interface for easy use

3. Technologies Used

- **Python**: Used as the main programming language
- **Flask**: A lightweight web framework used to build the backend
- **HTML, CSS, JavaScript**: Used for designing the frontend interface
- **VS Code**: Used as the development environment

4. Working of the System

The working of PlagioScan can be divided into several steps:

Step 1: User Input

The user enters two different texts into the input fields provided on the web page.

Step 2: Sending Data to Server

When the user clicks the check button, the data is sent to the Flask server using a POST request.

Step 3: Text Preprocessing

- Converts text to lowercase
- Removes punctuation
- Splits text into words (tokenization)
- Removes common words (stopwords)

This step ensures that only meaningful words are considered.

Step 4: TF-IDF Calculation

The system calculates TF-IDF (Term Frequency–Inverse Document Frequency) values for each word. This helps in identifying important words in the text.

Step 5: Similarity Calculation

Cosine similarity is used to measure how similar the two texts are. It compares the angle between two vectors representing the texts.

Step 6: Result Generation

Based on the similarity score:

- **High similarity** → Likely copied
- **Moderate similarity** → Partially similar
- **Low similarity** → Mostly original

Step 7: Display Output

The result is sent back to the frontend and displayed to the user in an easy-to-understand format.

5. Future Improvements

The project can be improved further by:

- Adding support for multiple documents
- Integrating with online plagiarism databases
- Using advanced NLP or AI models
- Improving UI design
- Adding file upload feature (PDF, Word)

6.How You Will Sell PlagioScan?

PlagioScan will be sold using a Freemium SaaS model — the same strategy used by Grammarly, Dropbox, and Slack.

- Free tier first — Users get 5 free checks per day with no credit card required. This removes all barriers to trying the product
- Website sign-up — Users visit plagioscan.com and start using it immediately
- Social media promotion — Target teacher communities, blogging groups, and student forums on Facebook, LinkedIn, and YouTube
- University outreach — Visit university IT departments and academic coordinators with live demos
- LMS integration — Partner with Moodle, Canvas, and Google Classroom as a plugin so institutions discover it naturally
- API marketplace — List on RapidAPI so developers find it and integrate it into their own platforms
- Word of mouth — One teacher who uses it tells 10 others. One institution adopts it and 500 students use it

7.Who Is Your Target Client?

There are 4 main client types:

1. Students & Teachers — Teachers want to check if students copied each other's short quiz answers or homework. Students want to verify their own work before submitting.

2. Academic Institutions — Universities and colleges that need to check thousands of short assignment submissions automatically across departments via LMS integration.

3. Content Writers & Bloggers — Freelance writers and marketers who need to make sure their blog posts, social captions, and ad copy are original before publishing — especially important for SEO.

4. E-Learning Platforms — Online quiz and course platforms that need an API to automatically check every student submission for plagiarism at scale.

8.Why a Client Would Purchase PlagioScan

1. No other tool does this — Turnitin, Grammarly, and Copyscape are all built for long documents. PlagioScan is the only tool designed specifically for short text (10–1000 words). Clients have no alternative.

2. It is extremely affordable — Turnitin charges $3–$5 per student per year for institutions only. Grammarly Premium costs $30/month. PlagioScan starts at just $9/month with unlimited checks for individuals.

3. Results in under 1 second — No file upload, no waiting, no complicated setup. Paste two texts and get the result instantly. Busy teachers and writers will not use slow tools.

4. Privacy guaranteed — Text is never stored after processing. Schools handling student data under GDPR or FERPA need this assurance. Competitors store submitted text in their databases.

5. API ready — E-learning platforms can integrate PlagioScan into their system in under 30 minutes using a simple REST API call. No competitor offers this at this price point.

6. Trusted algorithm — TF-IDF + Cosine Similarity is an academically validated method — not a black box. Teachers can explain and defend their decisions using the similarity score and the common keywords highlighted by the tool.

9. Conclusion

PlagioScan is a simple yet effective plagiarism detection system that demonstrates the use of Natural Language Processing techniques. It successfully compares two texts and provides meaningful insights about their similarity.

This project helped in understanding concepts like text preprocessing, TF-IDF, and cosine similarity. It also provided practical experience in building a full-stack application using Flask.

Overall, PlagioScan can be a useful tool for students and educators to ensure originality in written content.